CS 5402: Introduction to Data Mining

SP2026

(Lecture -4&5)

Similarity, Distance & Evaluation Basics

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Learning Objectives

By the end of this lecture, students will be able to:

- Compute and interpret common distance metrics
- Select similarity measures for different data types
- Explain train/validation/test splits
- Interpret confusion matrices and evaluation metrics

Why Do We Need Similarity/Distance Metrics?

Intuition:

Data mining often answers questions of "who is similar to whom" — and similarity must be quantified.

Without a distance (or similarity) metric:

- We cannot define "near" vs. "far"
- We cannot cluster, retrieve, or compare data points
- Many algorithms simply do not work

Customer	Annual Spend	Visits / Month
A	500	2
B	520	3
C	5,000	25

Customer Segmentation (Clustering):

Is A closer to B or to C?

Anomaly Detection (Outliers)

Problem

Detect fraudulent credit card transactions.

What is "Unusual behavior"?

What is a distance metric?

A function that measures how **different** two objects are

Properties:

- Non-negative
- Zero if identical
- Larger value → more different

Important:

Meaningfulness depends on **data type & domain**

Common Distance

L1 Distance
(Manhattan Distance)

$$d(x, y) = \sum |x_i - y_i|$$

Intuition:
- City blocks
- Robust to outliers

Common use:
- Sparse data
- High-dimensional data

L2 Distance
(Euclidean Distance)

$$d(x, y) = \sqrt{\sum (x_i - y_i)^2}$$

Intuition:
- Straight-line distance

Sensitive to:
- Feature scale
- Outliers

Requires: feature scaling

Cosine Similarity

$$\cos(x, y) = \frac{x \cdot y}{\|x\| \|y\|}$$

Measures:
- **Angle**, not magnitude
- **Range:** [-1, 1]

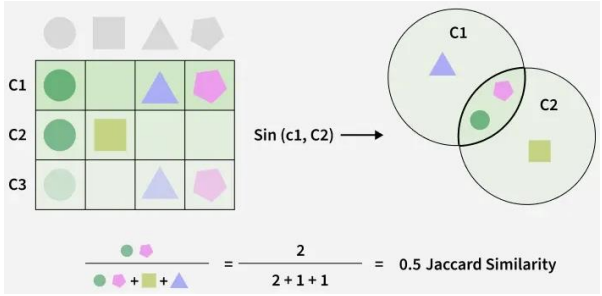
Common use:
- Text data
- High-dimensional vectors

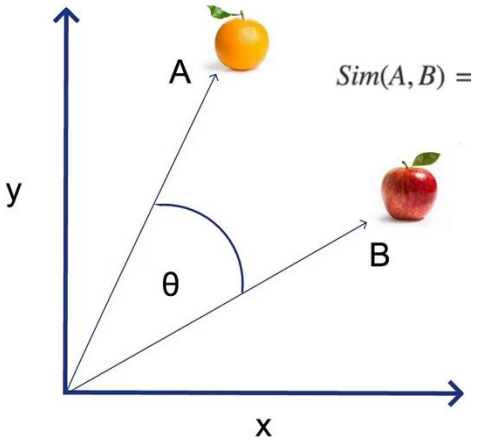
Jaccard Similarity

$$J(A, B) = \frac{|A \cap B|}{|A \cup B|}$$

Used for:
- Binary attributes
- Sets

Examples:
- Shopping baskets
- Keywords



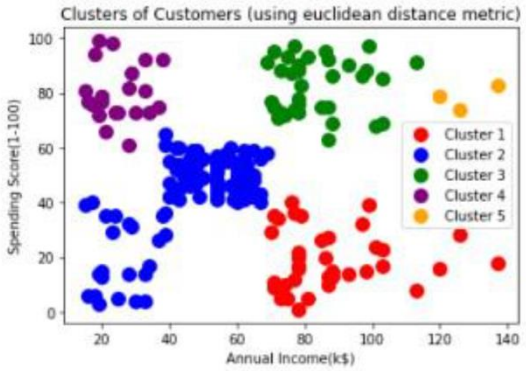


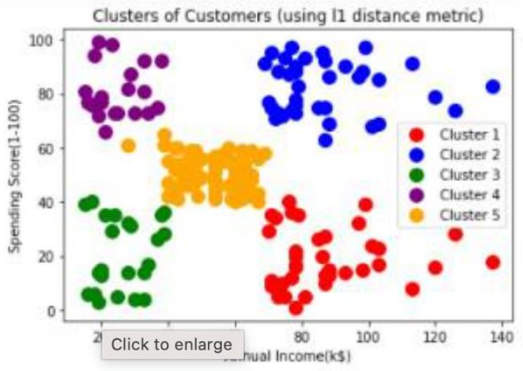
Similarity for Different Data Types

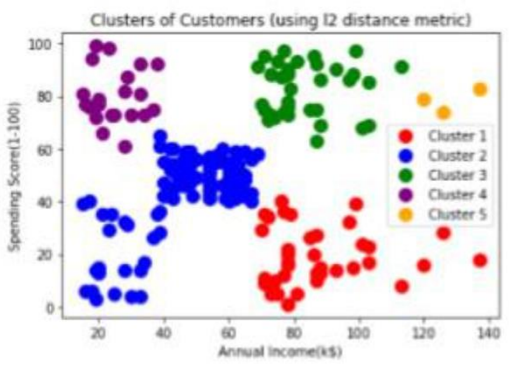
“Similarity” is not universal — it must respect how data is represented.
Using the wrong similarity measure can completely change clustering, classification, and retrieval results.

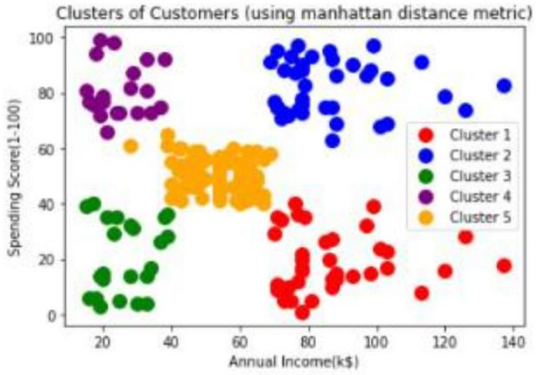
Data Type	Metric	Key Idea
Numeric	L1 / L2	Geometric distance
Binary (asymmetric)	Jaccard	Focus on presence
Categorical	Matching	Exact match
Ordinal	Rank + distance	Preserve order
Text	Cosine	Direction over magnitude
Set	Jaccard	Overlap
Mixed	Gower	Feature-wise similarity

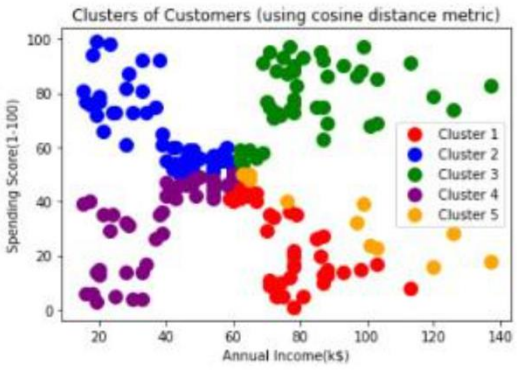
Example: Clustering with different metrics











https://www.geeksforgeeks.org/machine-learning/agglomerative-clustering-with-different-metrics-in-scikit-learn/

Why We Need Evaluation

1. To know how well it works
   - Reporting results
   - Making scientific/business decisions
   - Can we use it in real-life?



2. To compare models
   - For choosing which models are best
   - For tuning a given model's parameters



Model Evaluation

Model evaluation is the process of using different evaluation metrics to understand a machine learning model's performance, as well as its strengths and weaknesses.

Evaluating a model plays a vital role for the success of machine learning models.

- Holdout validation
- Cross validation

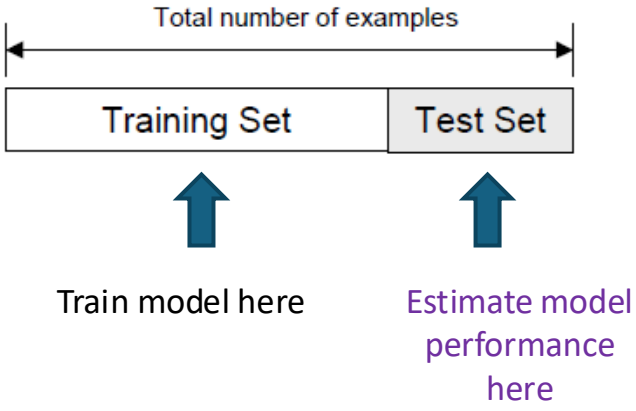
The basic rule in model validation

Don't test your model on data it's been trained with!

Holdout Validation

Holdout is the simplest approach. In this technique, the entire dataset is *randomly split* into:

- Training set: A dataset used for training the model
- Test set (a.k.a validation set): Data only used for testing the model



The three-way split

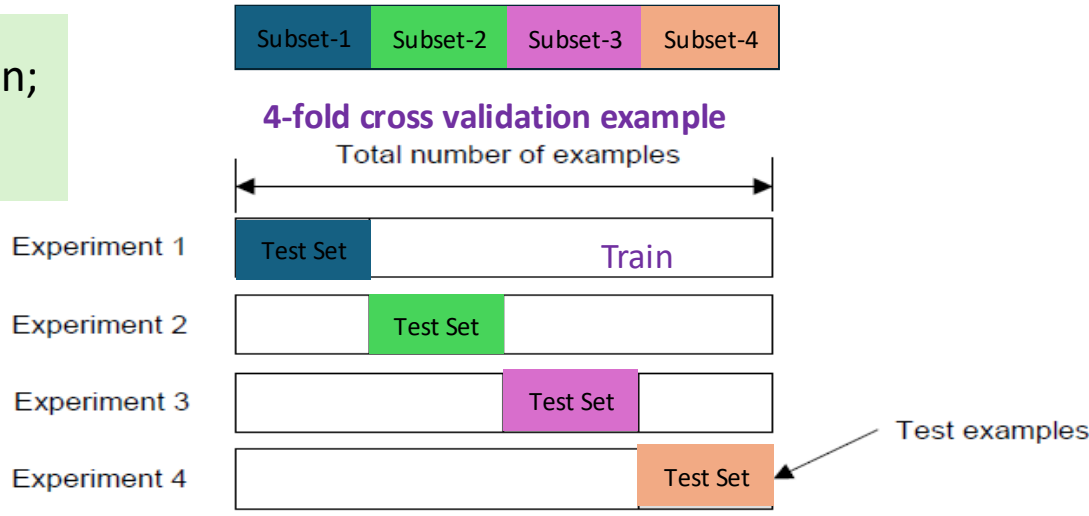
- Training set
  - A set of examples used for learning
- Validation set
  - A set of examples used to tune the parameters of a classifier
- Test set
  - A set of examples used only to assess the performance of fully-trained classifier.



Cross validation

- Cross validation is a method in which we do not use the whole dataset for training, in which, some part of the dataset is reserved for testing the model.
- There are many types of cross-validation, out of which K-fold cross-validation is mostly used.

- More reliable performance estimation;
- Better use of limited data.



Cross Validation (Cont'd)

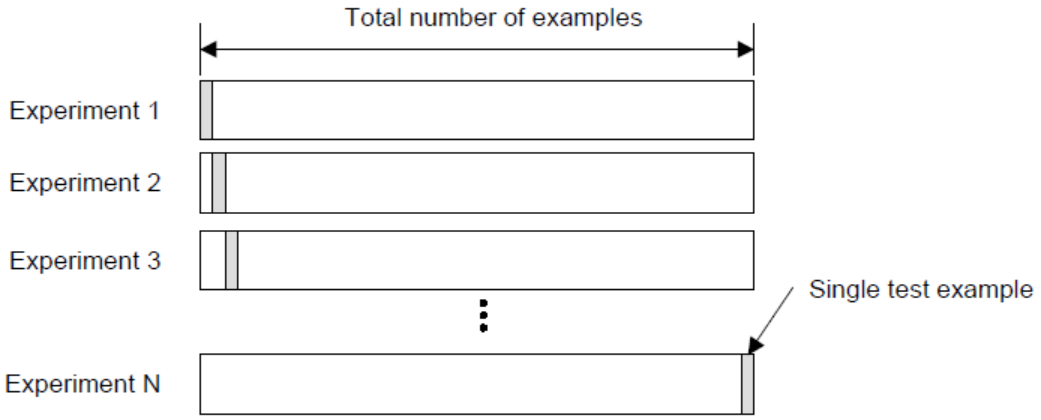
- How do you summarize the performance?
  - **Average**: Usually average of performance between experiments

$$E = \frac{1}{K}\sum_{i=1}^{K} E_i$$

- How many folds are needed?
  - Common choice: 5-fold or 10-fold cross validation
  - Large datasets → even a 3-Fold cross validation will be quite accurate
  - Smaller datasets → we will prefer bigger K
  - Leave-One-Out approach (K=n). In this case the number of folds K is equal to the number of examples n. Used for smaller datasets

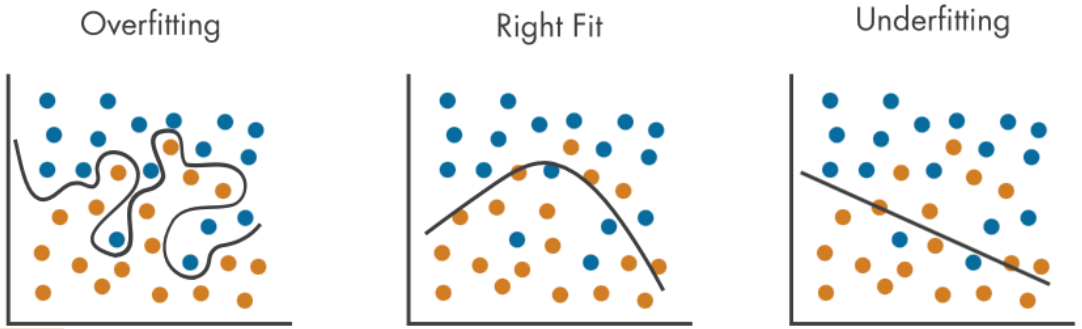
Leave-One-Out Cross Validation (LOOCV)

- Leave-One-Out is the degenerate case of K-Fold cross validation, where K is chosen as the total number of examples



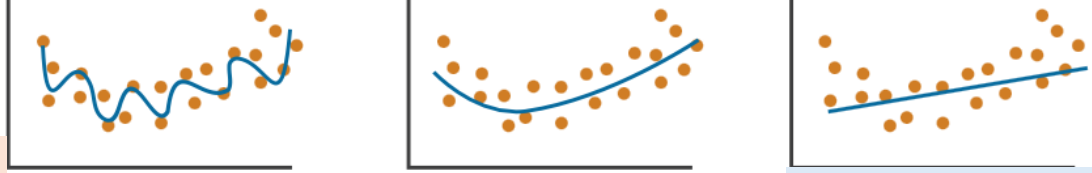
Overfitting vs. Underfitting

Both Overfitting and Underfitting refer to how well a model generalizes to unseen data.
They occur when a model is either too complex or too simple for the given dataset.



- Very low training error but high test error.
- Poor generalization to new, unseen data.
- The model performs well on training data but poorly on validation/test data

- High training and test errors.
- The model cannot capture important features.
- Poor performance even on training data.



- Too complex a model
- Insufficient training data
- Lack of regularization

- Model is too simple.
- Not enough training time or epochs
- Features are not properly engineered or missing key information

https://www.mathworks.com/discovery/overfitting.html

Example: The “99% Accuracy” Medical Paradox

- **The Scenario**: Imagine a very rare, but very serious, disease that only affects 1 in 1,000 people (0.1% of the population).
- **The "Lazy" Model:** I build a simple model that, no matter what, **always predicts "No Disease.”**

- **The "Amazing" Result:**
  - How often is this model right?
    - Out of 1,000 people, it correctly predicts "No Disease" for the 999 healthy people.
  - It only gets it wrong for the 1 person who is actually sick.
  - Its accuracy is **999 / 1,000 = 99.9%**!

- **The Punchline:** You have a 99.9% accurate model that would get a "pass" in most classes, but it completely fails at its one and only job: **finding the sick person.**

Example: The Spam Filter (**Precision** vs. Recall)

Scenario A: The Spam Filter

- Your model's job is to identify "Not Spam" (a "positive" result) and put it in the inbox.
- **Question:** What is the *worse* mistake?
  - **Mistake 1 (False Negative):** An important email (like a job offer) is missed and sent to the spam folder.
  - **Mistake 2 (False Positive):** A spam email (like a "buy-one-get-one" coupon) is marked as important and sent to your inbox.
- **The Answer:** Mistake 1 is *disastrous*. Mistake 2 is just *annoying*.
- **The Metric:** Your filter *must* be perfect when it says "this is important." It must have **high precision**. You're okay with letting some spam through (lower recall) if it means you *never* lose an important email.

Example: Doctor (Precision vs. **Recall**)

Scenario B: The Cancer Screener

- Your model's job is to identify patients who *might* have cancer (a "positive" result) for further testing.
- **Question:** What is the *worse* mistake?
  - **Mistake 1 (False Negative):** The model says a sick patient is "healthy," and they are sent home without follow-up.
  - **Mistake 2 (False Positive):** The model says a healthy patient "might be sick," and they are sent for a more expensive, but definitive, follow-up test (like a biopsy).
- **The Answer:** Mistake 1 is *catastrophic* and potentially fatal. Mistake 2 is stressful and costly, but not life-threatening.
- **The Metric:** Your model *must* find *every single person* who is sick. It must have **high recall**. You're willing to accept some false alarms if it means you *never* miss a real case.

Metrics Define Your Goal.

- "Are we failing to find sick patients?" → **Recall**
- "Are we sending job offers to spam?" → **Precision**
- "Is our model just 'lazy'?" → **Confusion Matrix / F1-Score**
- "Are we making a few *catastrophic* price guesses?" → **RMSE**

Choosing the right metric isn't just a math problem—it's the most important decision you'll make.

How do we evaluation models?

What is the criteria for a good ML model, or otherwise?

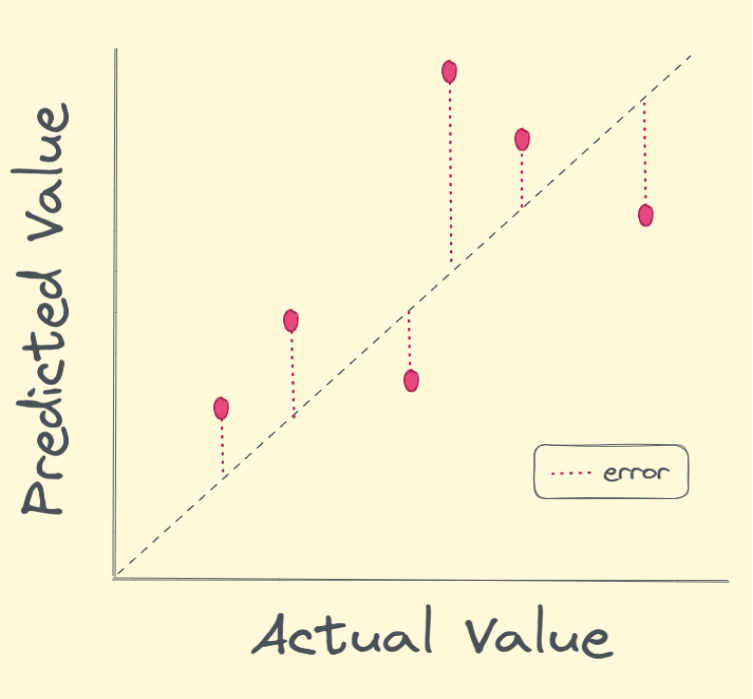
- Criteria (Application-dependent):
  - Accuracy
  - Misclassification error, or risk (loss functions)
  - Training time/space complexity
  - Testing time/space complexity
  - Interpretability (Model Complexity, for instance number of hidden units)
  - Less computational cost
  - Easy tuning
  - Easy implementation
  - etc

Performance metrics (Regression)

- For regression model:

$$e_t = y_i - \hat{y}_i$$

Mean squared error	$\text{MSE} = \frac{1}{n}\sum_{t=1}^{n} e_t^2$
Root mean squared error	$\text{RMSE} = \sqrt{\frac{1}{n}\sum_{t=1}^{n} e_t^2}$
Mean absolute error	$\text{MAE} = \frac{1}{n}\sum_{t=1}^{n} \lvert e_t \rvert$
Mean absolute percentage error	$\text{MAPE} = \frac{100\%}{n}\sum_{t=1}^{n} \left\lvert \frac{e_t}{y_t} \right\rvert$

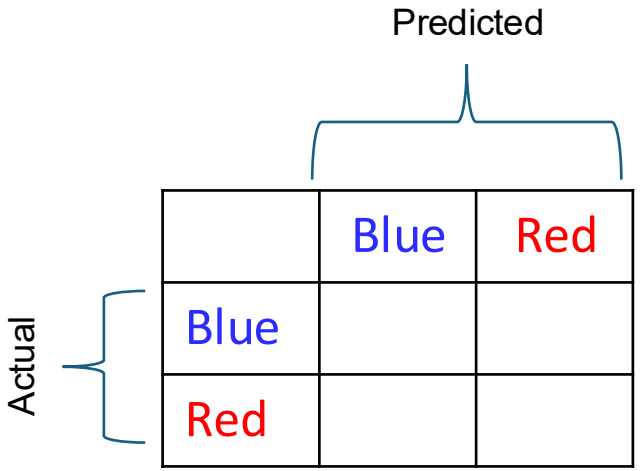


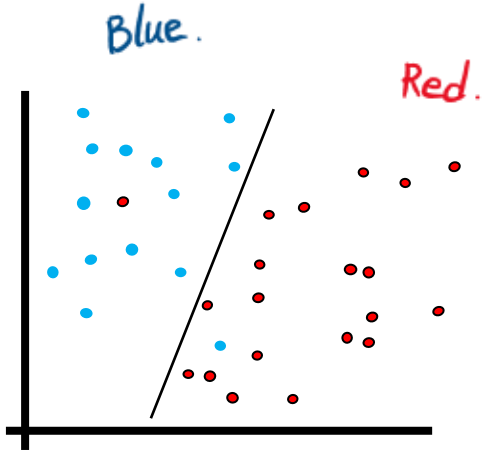
Performance metrics (Classification)

- For classification task:
  - **Accuracy**
    - Closeness to true values
    - how often is a classifier correct overall?
  - **Precision & Recall (and f1 score)**
    - Precision: What percent of our predictions are accurate?
    - Recall: How many of the accurate predictions did we capture
    - F1 score: A single number that combines the two values above. Good for ranking/sorting…
  - **Specificity/sensitivity**
    - very similar to precision/recall. Often used in medicine.
  - **Area under curve (the ROC curve)**
    - A Receiver Characteristic Curve (ROC) plots the True positive rate (TPR) vs. the False positive rate (FPR). The maximum area under the curve (AUC) is 1. Completely random predictions have an AUC of 0.5. The advantage of this metric is that it is continuous.

Confusion Matrix (1)

Confusion Matrix = table to visualize the performance of a classifier

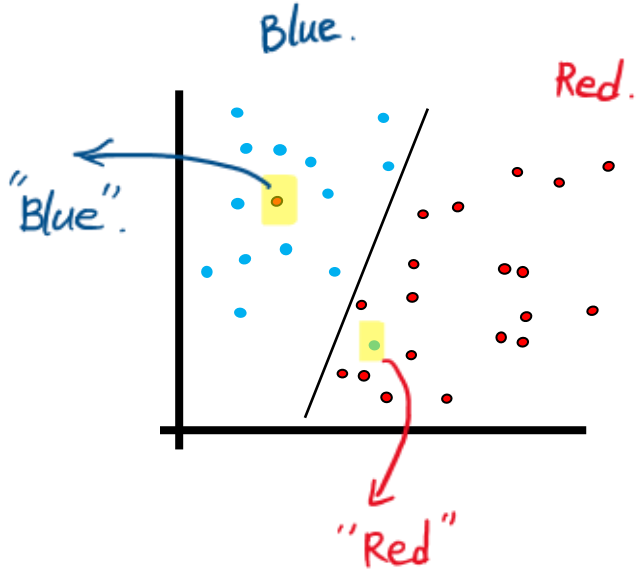




Confusion Matrix (2)

Predicted

Actual		Blue	Red
	Blue	13	1
	Red	1	19



Confusion Matrix (3)

		Predicted	
		0 (NEGATIVE)	**1 (POSITIVE)**
Actual	**0 (NEGATIVE)**	True Negative (**TN**) (Correctly rejected!)	False Positive (**FP**) (A false alarm!) Type I Error
	1 (Positive)	False Negative (**FN**) (A miss!) Type II Error	True Positive (**TP**) (Correctly found!)

Accuracy

Accuracy, the "Overall Correct" metric.

Question: what percentage of all predictions were right?

Formula: $\frac{TN+TP}{TN+FP+FN+TP}$

		Predicted	
		0 (Healthy)	1 (Sick)
Actual	**0** (Healthy)	True Negative (**TN**) 85	False Positive (**FP**) 5
	1(Sick)	False Negative (**FN**) 2	True Positive (**TP**) 8

Example: $\frac{85+8}{85+5+2+8} = \frac{93}{100} = 93\%$

Accuracy is a good starting point, but it's very misleading on imbalanced datasets (where one class is much more common than the other)

Precision

Precision, the "Purity" metric.

Question: Of all the times we predicted positive, how often were we correct?

Focus on the "Predicted Positive" column

Formula: $\frac{TP}{FP+TP}$

Example: $\frac{8}{5+8} = \frac{8}{13} = 61.5\%$

		Predicted	
		0 (Healthy)	1 (Sick)
Actual	0 (Healthy)	True Negative (**TN**) 85	False Positive (**FP**) 5 ★
	1(Sick)	False Negative (**FN**) 2	True Positive (**TP**) 8

- **Use Case:** Use when the cost of a **False Positive** is high.
  - **Spam Filter:** You *hate* it when a good email (Actual Negative) goes to spam (Predicted Positive). You want your inbox predictions to be *precise*.

Recall

Recall, the "Completeness" metric.

Question: Of all the actual positive cases, what fraction did we find?

Focus on the "Actual Positive" row.

Formula: $\frac{TP}{FN+TP}$

Example: $\frac{8}{2+8} = \frac{8}{10} = 80\%$

		Predicted	
		0 (Healthy)	1 (Sick)
Actual	**0** (Healthy)	True Negative (**TN**) 85	False Positive (**FP**) 5
	1(Sick)	False Negative (**FN**) 2 ★	True Positive (**TP**) 8

- **Use Case:** Use when the cost of a **False Negative** is high.
  - **Cancer Test:** You *must* find every sick patient. A missed case (a False Negative) is a disaster. You don't mind a few false alarms (False Positives) if it means you find all the real cases.

Quick Question: Which Metric Matters More?

For each scenario, would you optimize for **Precision** or **Recall**? Why?

- **Airport Security Screening:** Your model flags bags for a manual check to find weapons. (Positive = "Weapon").
  - *What's the worse error?* A False Positive (clean bag is checked) or a False Negative (weapon is missed)?
- **Bank Fraud Detection:** Your model flags transactions as "fraudulent" and blocks the credit card. (Positive = "Fraud").
  - *What's the worse error?* A False Positive (you block a real purchase) or a False Negative (you let a fraudulent charge go through)?
- **Email Marketing:** Your model predicts which customers will *actually* buy from a coupon. (Positive = "Will Buy").
  - *What's the worse error?* A False Positive (you send a coupon to someone who won't buy) or a False Negative (you *don't* send a coupon to someone who *would* have bought)?

F1-Score

F1-Score, the "Balancer" metric.

Question: How can we get a single number that balances Precision and Recall?

Answer: The Harmonic Mean. It's a "better" average that heavily punishes extreme, low values.

Formula: $2 * \frac{Precision * Recall}{Precision + Recall}$

Example: $2 * \frac{0.615 * 0.8}{0.615 + 0.8} = \frac{0.984}{1.415} = 69.5\%$

		Predicted	
		0 (Healthy)	1 (Sick)
Actual	0 (Healthy)	True Negative (TN) 85	False Positive (FP) 5 ★
	1(Sick)	False Negative (FN) 2 ★	True Positive (TP) 8

- **Use Case:** A great "all-in-one" metric, especially when your classes are imbalanced and you care about *both* kinds of errors.

Question-1:

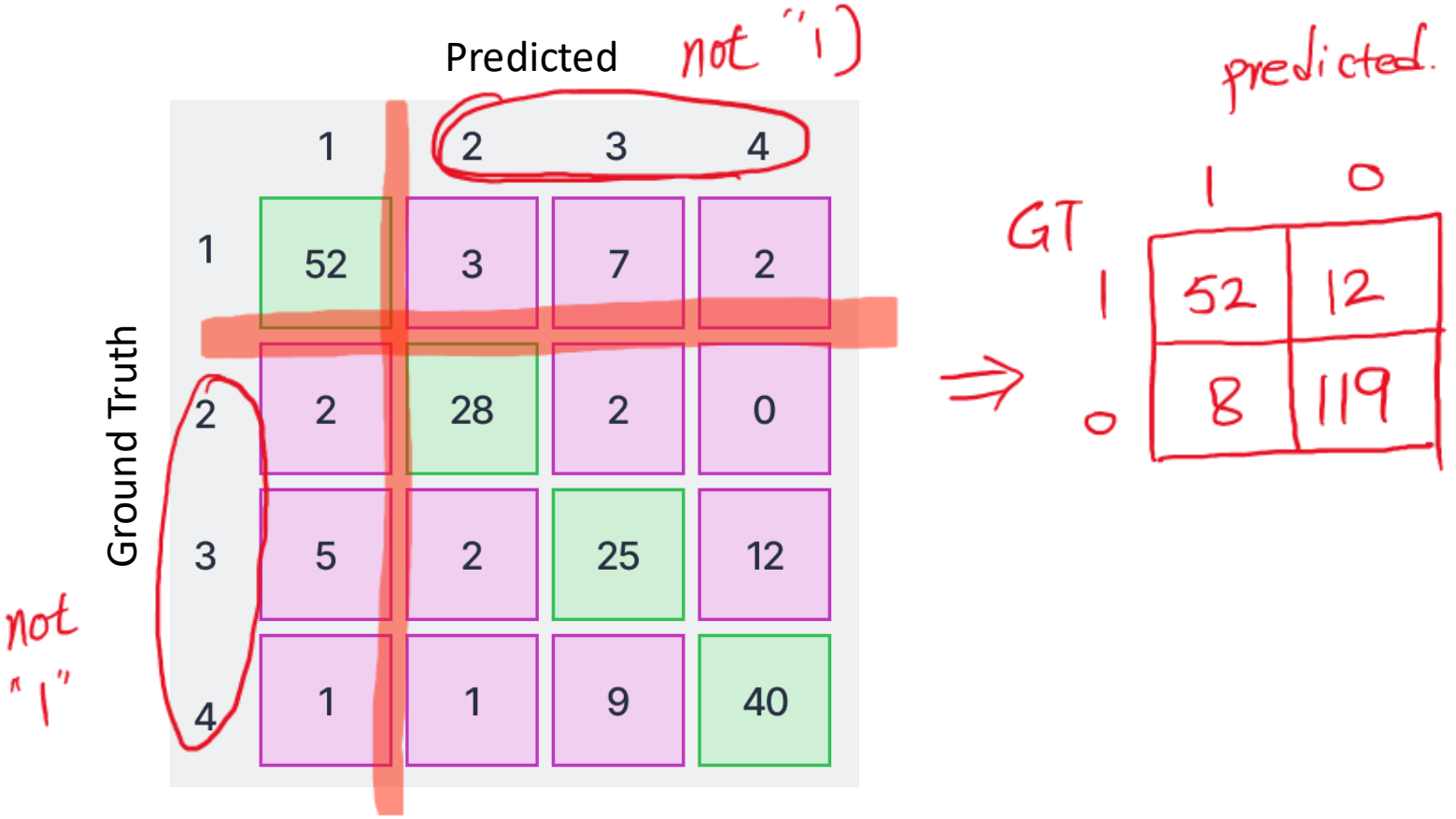
		Predicted	
		0	1
Actual	0	True Negative (TN) 170	False Positive (FP) 70
	1	False Negative (FN) 40	True Positive (TP) 120

- Recall = $\frac{TP}{FN+TP}$
- Precision = $\frac{TP}{FP+TP}$
- F-Measure = $2 * \frac{Precision*Recall}{Precision+Recall}$

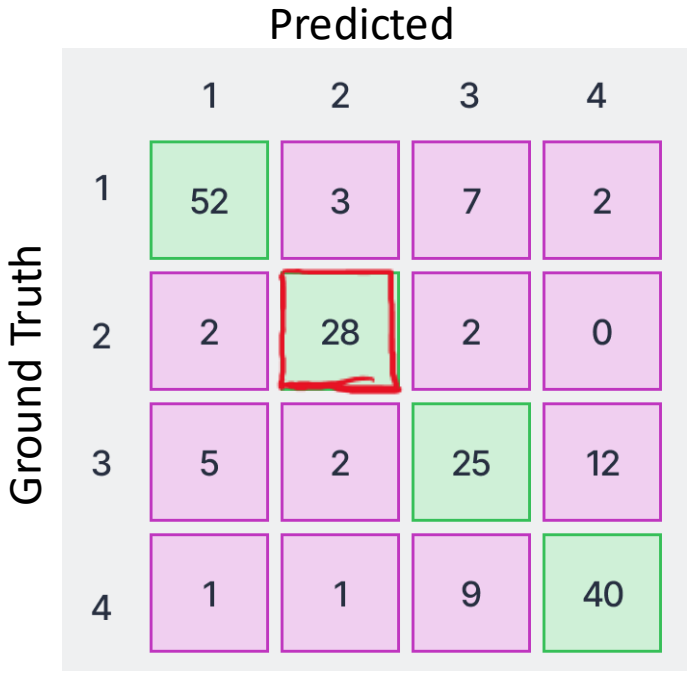
what is F1 score?

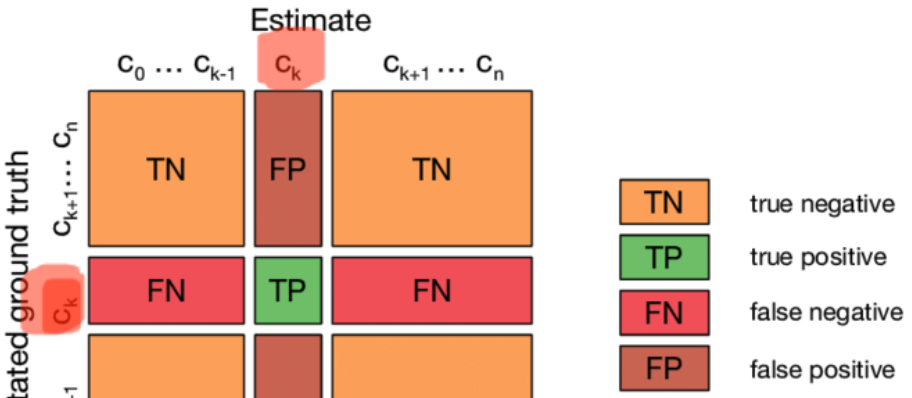
A. 0.6315
B. 0.75
C. 0.6857
D. 0.725

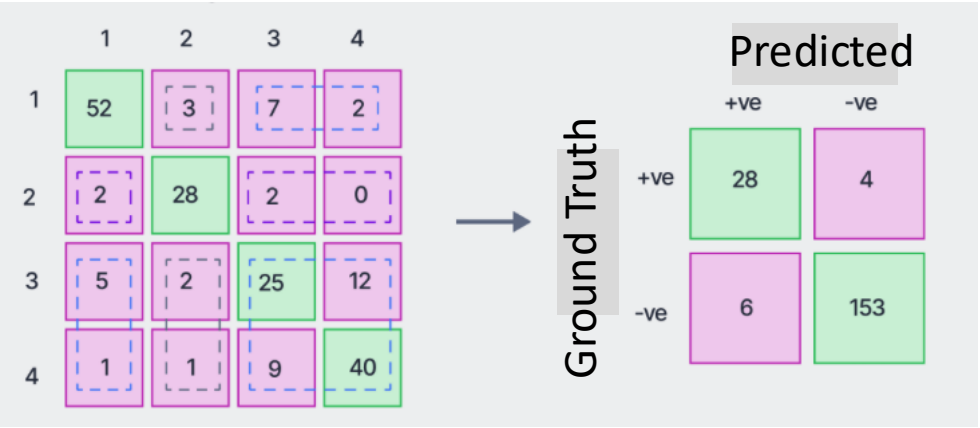
Confusion matrix with multiple classes (1)



Confusion matrix with multiple classes (2)





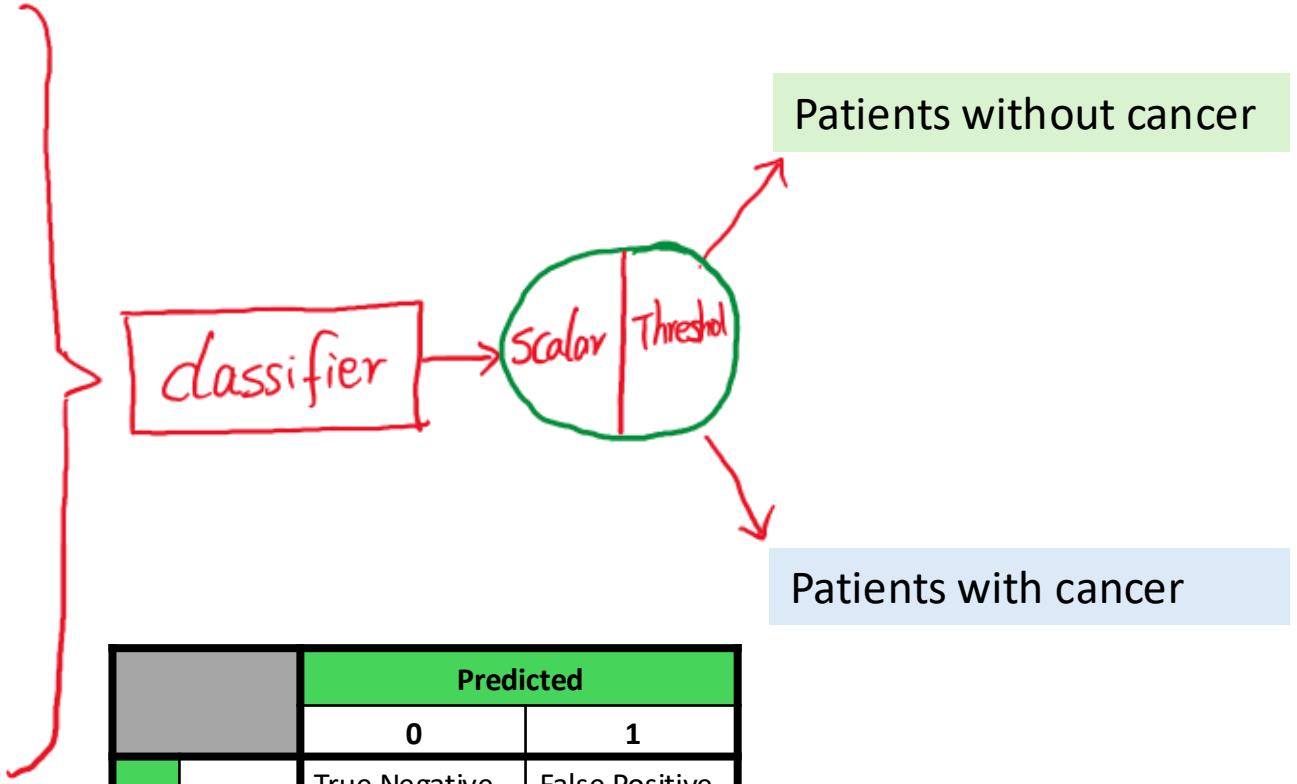


Beyond Yes/No: Probabilities & Thresholds

Let's think about how classifiers distinguish classes?







		Predicted	
		0	1
Actual	0	True Negative 100	False Positive **0**
	1	False Negative 0	True Positive 100

How can we evaluate the model across all possible thresholds at once?

Most models don't just say "Yes" or "No." They give a probability score.

- Email A: 95% chance of spam
- Email B: 60% chance of spam
- Email C: 15% chance of spam

We choose a threshold to make the final decision.

- Threshold = 50%: A and B are "Spam." C is "Good."
- Threshold = 80%: Only A is "Spam." B and C are "Good."

Changing the threshold changes our Precision and Recall!

- High Threshold (e.g., 90%): High Precision, Low Recall.
- Low Threshold (e.g., 30%): Low Precision, High Recall.

How can we evaluate the model across all possible thresholds at once?

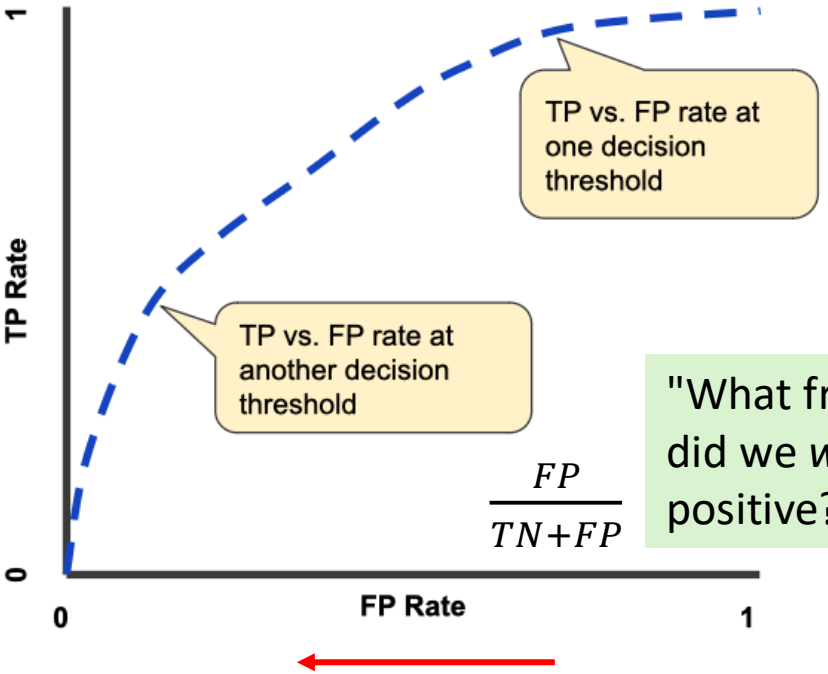
ROC Curve

An ROC curve (*receiver operating characteristic curve*) is a graph showing the performance of a classification model at all classification thresholds.

This curve plots two parameters:

- True Positive Rate (TPR)
- False Positive Rate (FPR)

		Predicted	
		0 (NEGATIVE)	1 (POSITIVE)
Actual	0 (NEGATIVE)	True Negative (TN) (Correctly rejected!)	False Positive (FP) (A false alarm!) Type I Error
	1 (Positive)	False Negative (FN) (A miss!) Type II Error	True Positive (TP) (Correctly found!)



(Recall)

$$\frac{TP}{FN+TP}$$

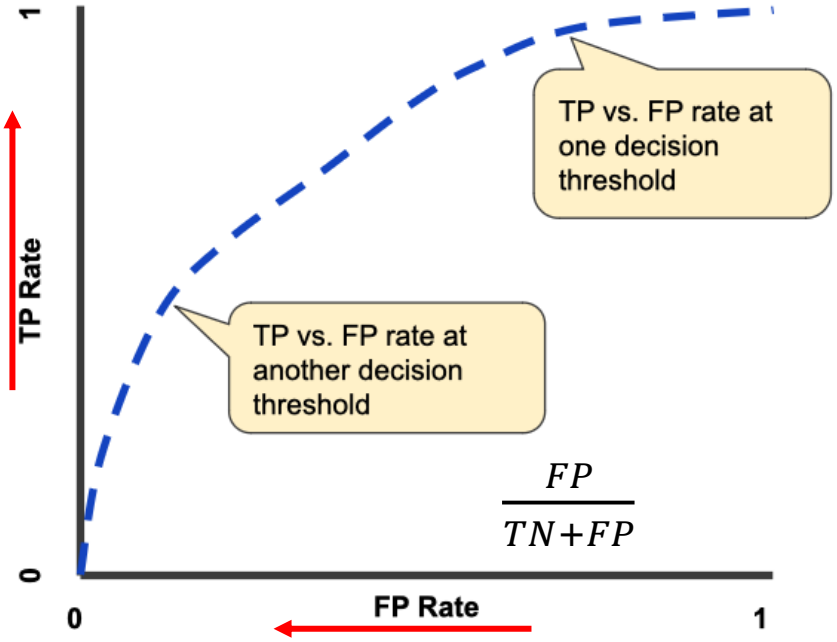
$$\frac{FP}{TN+FP}$$

"What fraction of *negatives* did we *wrongly* flag as positive?" (Want this to be 0)

How to Read a ROC Curve (1)

(Recall)

$$\frac{TP}{FN+TP}$$



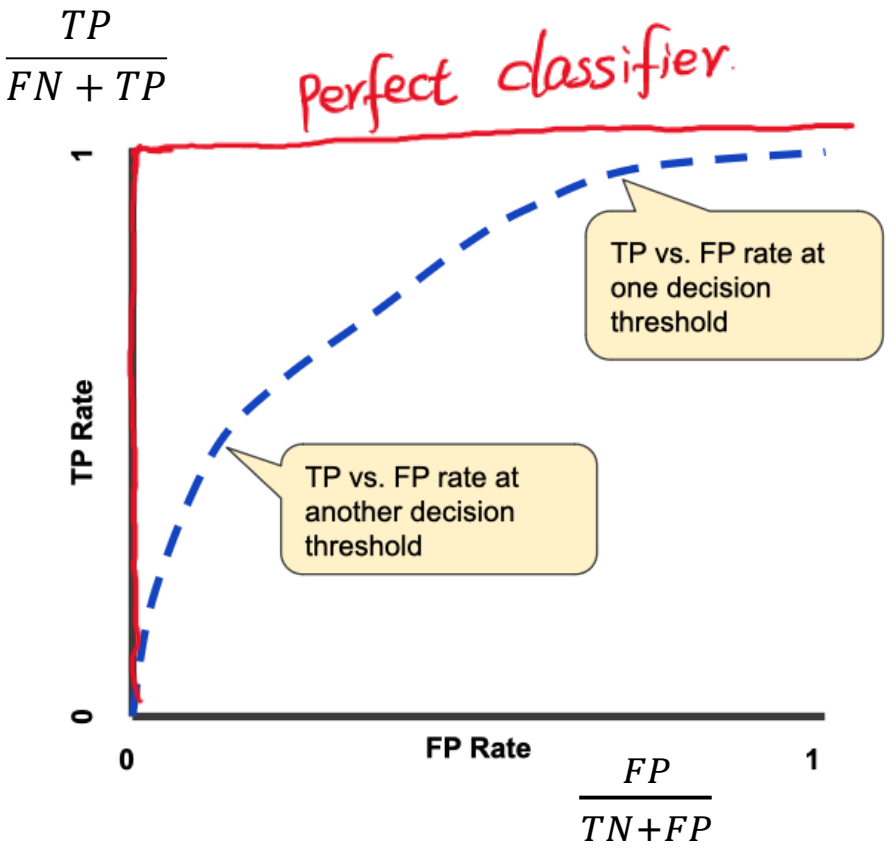
		Predicted	
		0 (NEGATIVE)	1 (POSITIVE)
Actual	0 (NEGATIVE)	True Negative (TN) (Correctly rejected!)	False Positive (FP) (A false alarm!) Type I Error
	1 (Positive)	False Negative (FN) (A miss!) Type II Error	True Positive (TP) (Correctly found!)

- **Point (0, 0):** Threshold = 1.0 (100%).
  - Predicts "Negative" for everything. (TPR=0, FPR=0).
- **Point (1, 1):** Threshold = 0.0 (0%).
  - Predicts "Positive" for everything. (TPR=1, FPR=1).

How to Read a ROC Curve (2)

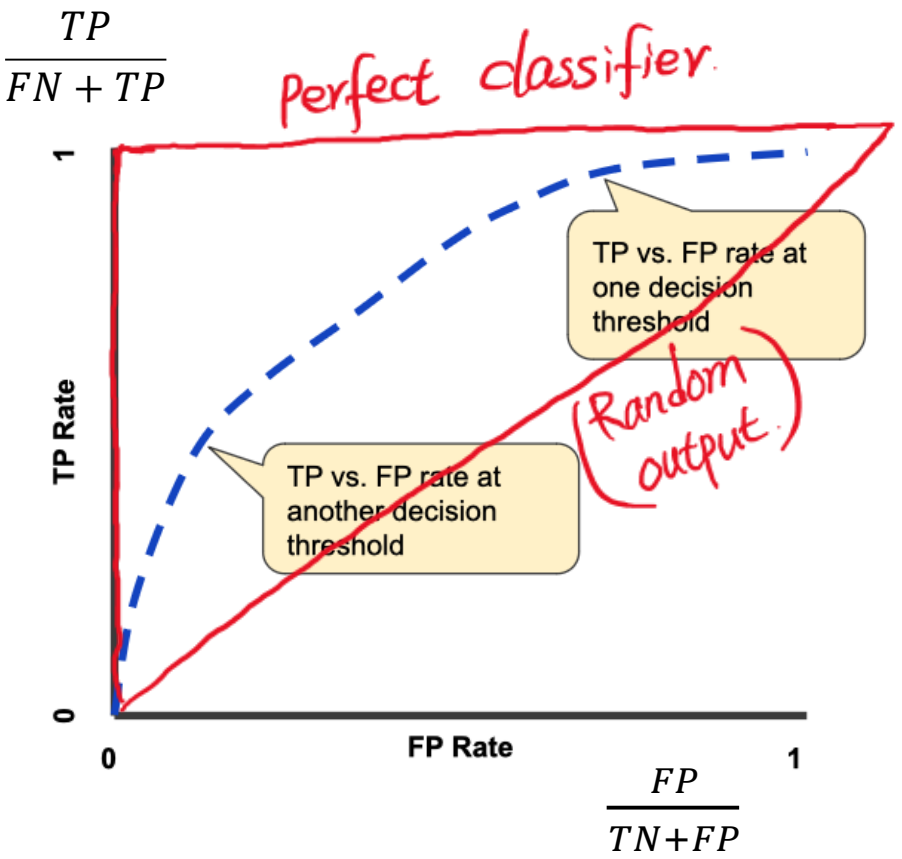
		Predicted	
		0	1
Actual	0	True Negative (TN) #	False Positive (FP) O
	1	False Negative (FN) O	True Positive (TP) #

perfect classifier: for whatever threshold: FP = 0. TP = 1



How to Read a ROC Curve (3)

		Predicted	
		0	**1**
Actual	**0**	True Negative (**TN**)	False Positive (**FP**)
	1	False Negative (**FN**)	True Positive (**TP**)



How to draw ROC Curve? (1)

developed a model that provides score
higher score -> label = 1

X	Y (ground truth)	Y' (predicted)
X1	1	0.95
X2	1	0.92
X3	0	0.80
X4	1	0.76
x5	1	0.71

How to draw ROC Curve? (2)

X	Y (ground truth)	Y' (predicted)	Y~ (t=0.95)
X1	1	0.95	1
X2	1	0.92	0
X3	0	0.80	0
X4	1	0.76	0
x5	1	0.71	0

Thresholding=0.95

TPR (t=0.95)
FPR (t=0.95)

		Predicted	
		0	**1**
Actual	**0**	True Negative (**TN**)	False Positive (**FP**)
	1	False Negative (**FN**)	True Positive (**TP**)

How to draw ROC Curve? (3)

X	Y (ground truth)	Y' (predicted)	Y~ (t=0.95)	Y~ (t=0.92)
X1	1	0.95	1	1
X2	1	0.92	0	1
X3	0	0.80	0	0
X4	1	0.76	0	0
x5	1	0.71	0	0

Thresholding=0.92

TPR (t=0.92)
FPR (t=0.92)

How to draw ROC Curve? (4)

X	Y (ground truth)	Y' (predicted)	Y~ (t=0.95)	Y~ (t=0.92)	Y~ (t=0.80)	Y~ (t=0.76)	Y~ (t=0.71)
X1	1	0.95	1	1	1	1	1
X2	1	0.92	0	1	1	1	1
X3	0	0.80	0	0	1	1	1
X4	1	0.76	0	0	0	1	1
x5	1	0.71	0	0	0	0	1

Thresholding=0.80, 0.76, 0.71

TPR (t=0.80) , TPR (t=0.76) , TPR (t=0.71)

FPR (t=0.80) , FPR(t=0.76) , FPR(t=0.71)

Draw ROC curve with Pairs of TPR and FPR

- Using N thresholds (N samples)

We will get pairs of TPR and FPR for each N

TPR, FPR for threshold 1

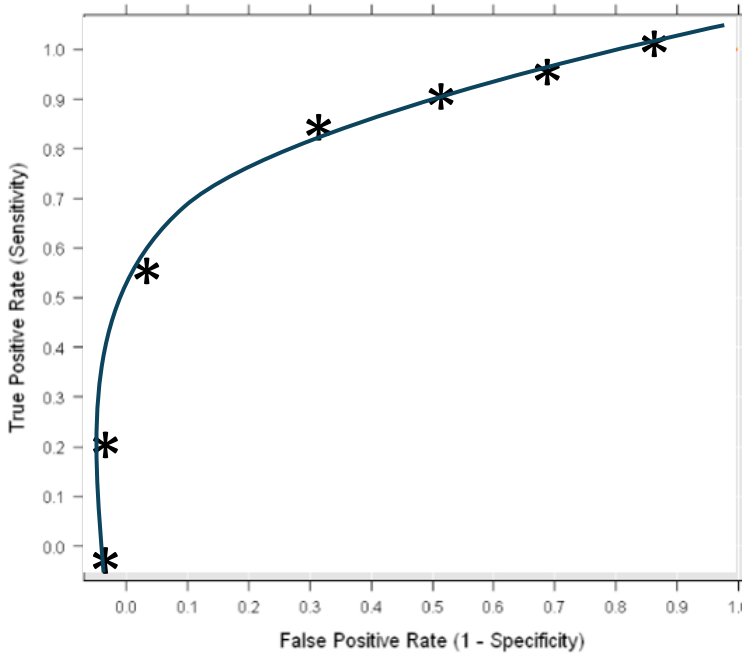
TPR, FPR for threshold 2

TPR, FPR for threshold 3

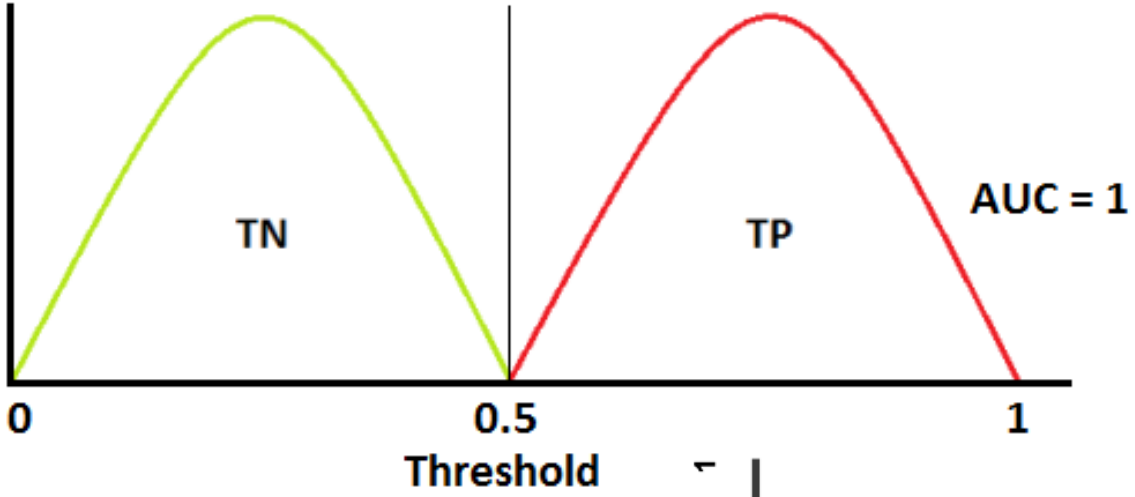
TPR, FPR for threshold 4

….

TPR, FPR for threshold N

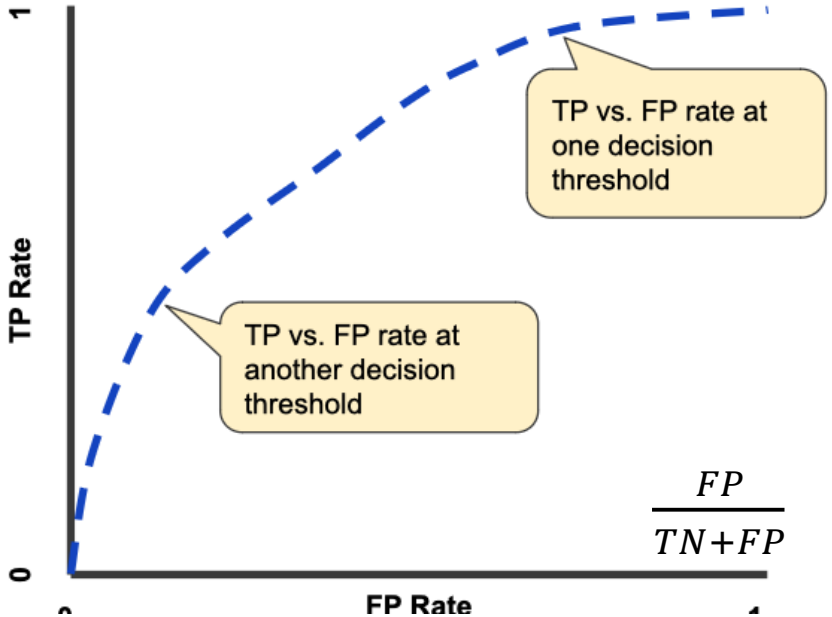


Example: Scenario-1 (ideal case)

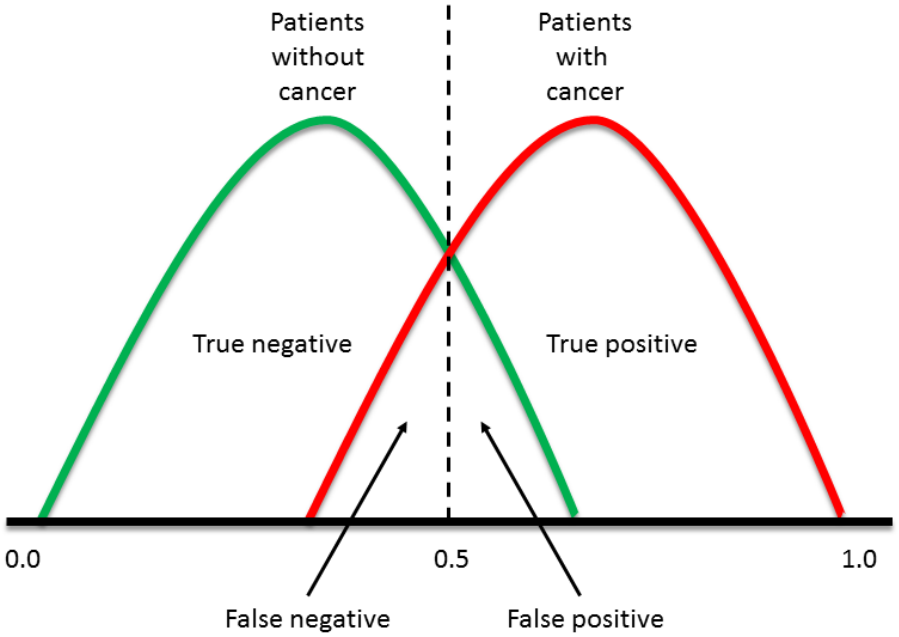


		Predicted	
		0	**1**
Actual	**0**	True Negative (**TN**)	False Positive (**FP**)
	1	False Negative (**FN**)	True Positive (**TP**)

$$\frac{TP}{FN+TP}$$

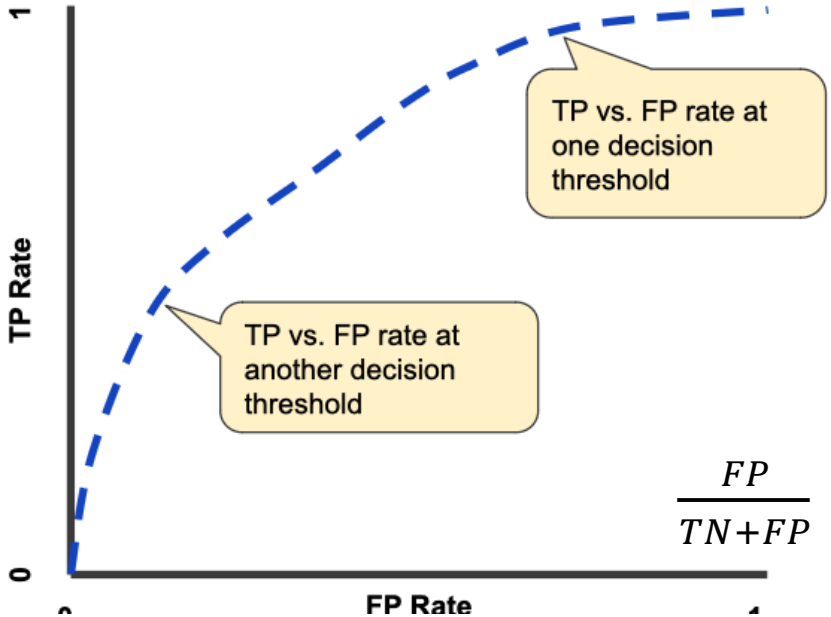


Example: Scenario-2

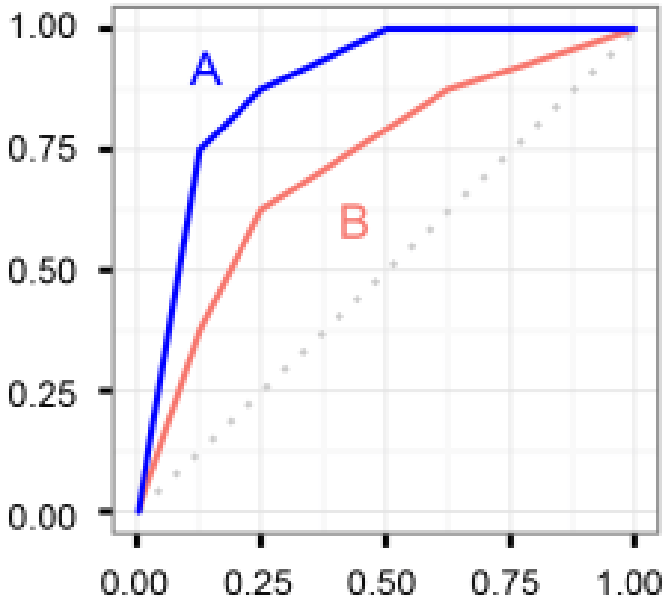


		Predicted	
		0	**1**
Actual	**0**	True Negative (**TN**)	False Positive (**FP**)
	1	False Negative (**FN**)	True Positive (**TP**)

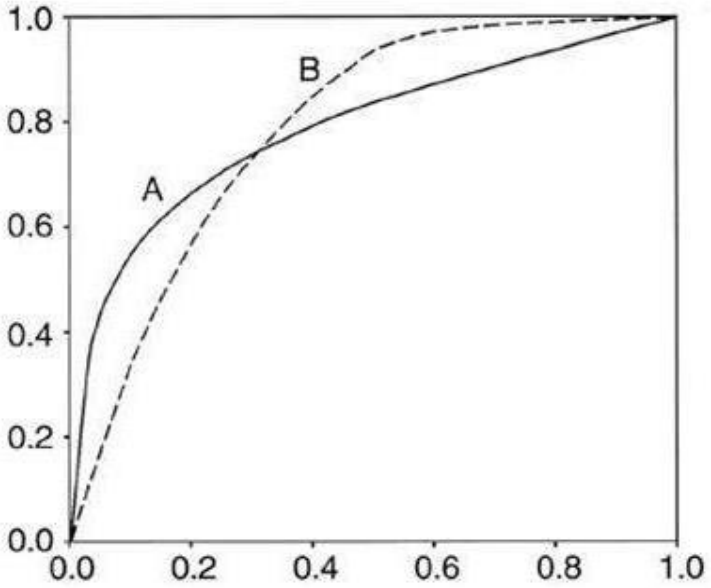
$$\frac{TP}{FN+TP}$$



How do we compare two models on a ROC graph?



VS.



AUC: the “One-Number” Summary

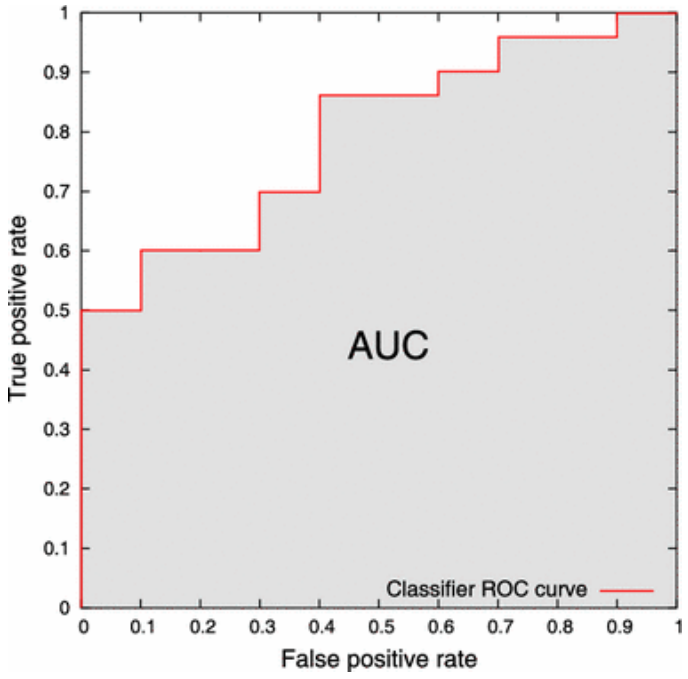
How do we compare two models (e.g., Model A vs. Model B) on an ROC graph?

AUC stands for “Area under the ROC Curve”, which measures the entire two-dimensional area underneath the entire ROC curve.

AUC provides an aggregate measure of performance across all possible classification thresholds.

Interpreting AUC:

- AUC = 1.0: Perfect model.
- AUC = 0.8 - 0.99: Excellent model.
- AUC = 0.7 - 0.8: Good model.
- AUC = 0.5: Useless "random guess" model.
- AUC < 0.5: Your model is *worse* than random! (Tip: Just flip its predictions).



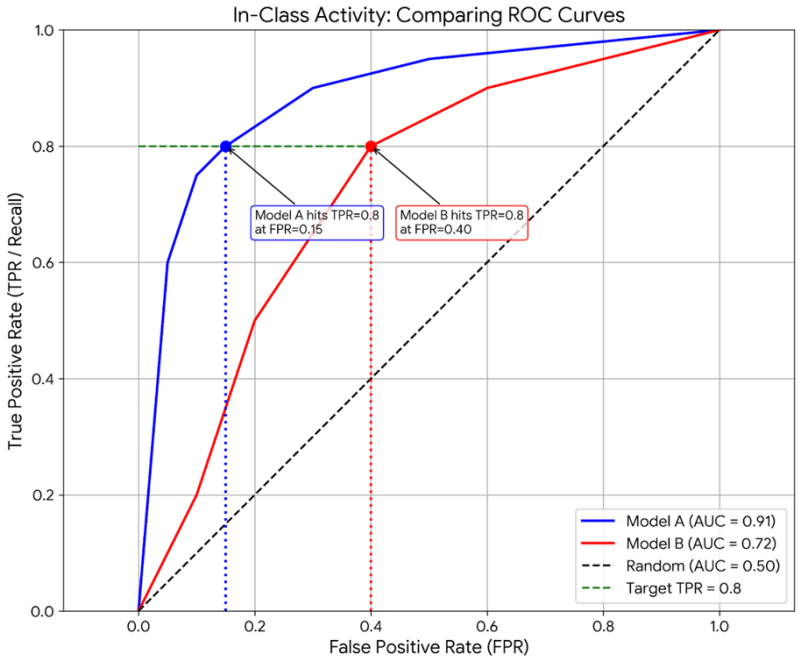
AUC is one of the most-used metrics for "overall" model quality because it's threshold-independent.

Questions for Further Thought (1)

- Can ROC handle imbalanced datasets?

Questions for Further Thought (2)

- Your team has built two models to predict customer churn.
- **Which model is better overall?**
- **Challenge Question:** Your boss says, "I don't care about false alarms. I want to find *at least 80%* of the customers who are *actually* going to leave (TPR/Recall = 0.8)." Which model is better *for this specific task*?



Evaluation Considerations

- Higher accuracy does not necessarily imply better performance on target task

- Classes are often very imbalanced
  - Buying customers are much rarer than non buying customers
  - What if 99:1 split? Always say “no” -- 99% default accuracy

- Errors have different ***costs***
  - Example: Cost of eating poisonous mushroom is greater than cost of going hungry

Summary: Your Evaluation Toolkit

Metric	Question It Answers	When to Use It
Accuracy	"Overall, how often is the model right?"	Simple problems. **Warning:** Misleading on imbalanced data.
Confusion Matrix	"Where is the model getting confused?"	Always! Start here.
Precision	"When it predicts positive, how often is it right?"	When **False Positives** are costly (e.g., spam filter, legal doc review).
Recall	"Of the actual positives, how many did it find?"	When **False Negatives** are costly (e.g., cancer test, security).
F1-Score	"What's the balanced score between P & R?"	Great all-rounder for imbalanced classes.
ROC / AUC	"How good is the model at all thresholds?"	Excellent for comparing models "apples-to-apples."